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CPCS-214 Computer Architecture and Organization

22 - Performance

Introduction

- Assessing performance of computers can be quite challenging
- Scale and intricacy of modern SW systems, together with wide range of performance improvement techniques employed by HW designers, have made performance assessment much more difficult
- Will describe metrics for measuring performance

Defining Performance

- When we say one computer has better performance than another, what do we mean?
- Although this question might seem simple
  - Analogy with passenger airplanes shows how subtle the question of performance can be

Defining Performance

- Which of the planes had best performance?

Airplane	Passenger capacity	Cruising range (miles)	Cruising speed (m.p.h.)	Passenger throughput (passengers x m.p.h.)
Boeing 777	375	4630	610	228,750
Boeing 747	470	4150	610	286,700
BAC/Sud Concorde	132	4000	1350	178,200
Douglas DC-8-50	146	8720	544	79,424

Defining Performance

- Which of the planes had best performance?
- Need to define performance; Plane with
  - highest cruising speed is Concorde
  - longest range is DC-8
  - largest capacity is 747

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Defining Performance

- Define fastest plane as one with highest cruising speed
  - Taking single passenger from one point to another in least time
    - Transporting 450 passengers

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Defining Performance

- Define fastest plane as one with highest cruising speed
  - Taking single passenger from one point to another in least time
    - Transporting 450 passengers
      - 747 would be the fastest

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Computer Performance

- Similarly, we can define computer performance in several different ways
- Running a program on two different desktop computers
  – Faster one is the computer that gets job done first
- Datacenter that had several servers running jobs
  – Faster is the one that completed most jobs during a day

Execution Time and Throughput

- Computer user is interested in reducing execution time
  - Time between start and completion of a task
- Datacenter managers are interested in increasing throughput or bandwidth
  - Total amount of work done in a given time

Performance

- In discussing performance of computers, we will be primarily concerned with execution time
- To maximize performance, we want to minimize execution time for some task
- We can relate performance and execution time for computer X:

$$\text{Performance}_X = \frac{1}{\text{Execution time}_X}$$

Performance

- For two computers X and Y, if performance of X is greater than performance of Y, we have

$$\frac{\text{Performance}_X}{\text{Performance}_Y} = n$$

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$$\frac{\text{Performance}_X}{\text{Performance}_Y} = n$$

- If X is *n* times as fast as Y, then execution time on Y is *n* times as long as it is on X:

$$\frac{\text{Performance}_X}{\text{Performance}_Y} = \frac{\text{Execution time}_Y}{\text{Execution time}_X} = n$$

Example

- Computer A runs a program in 10 seconds and computer B runs same program in 15 seconds, how much faster A than B?

Example (cont.)

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- A is *n* times as fast as B if

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Example (cont.)

- Computer A runs a program in 10 seconds and computer B runs same program in 15 seconds, how much faster A than B?
- A is *n* times as fast as B if

$$\frac{\text{Performance}_A}{\text{Performance}_B} = \frac{\text{Execution time}_B}{\text{Execution time}_A} = n$$

- Performance ratio is $$\frac{15}{10} = 1.5$$

A is therefore 1.5 times as fast as B

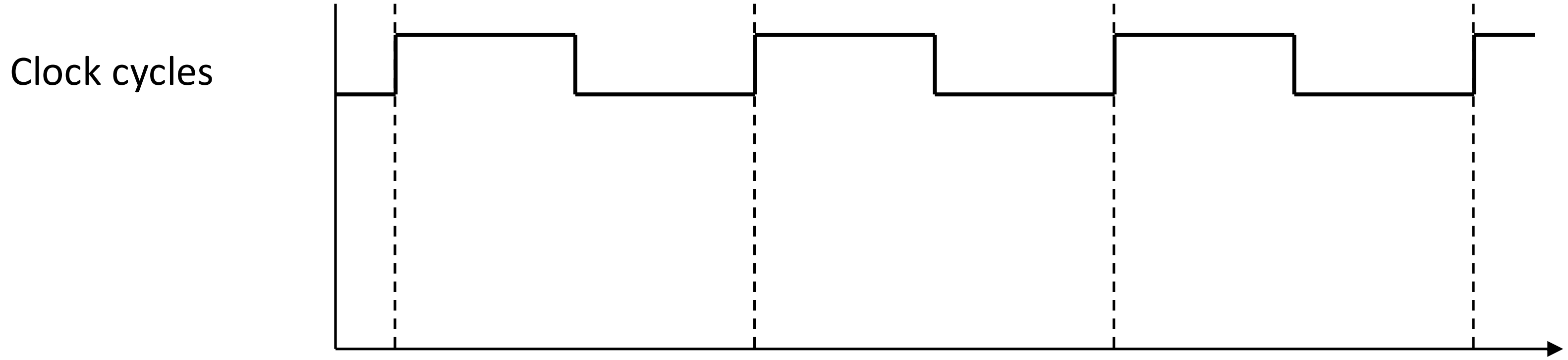
- Could also say that computer B is 1.5 times *slower than* computer A

Measuring Performance

- Time is the measure of computer performance: computer that performs same amount of work in the least time is the fastest
- Program *execution time* is measured in seconds per program

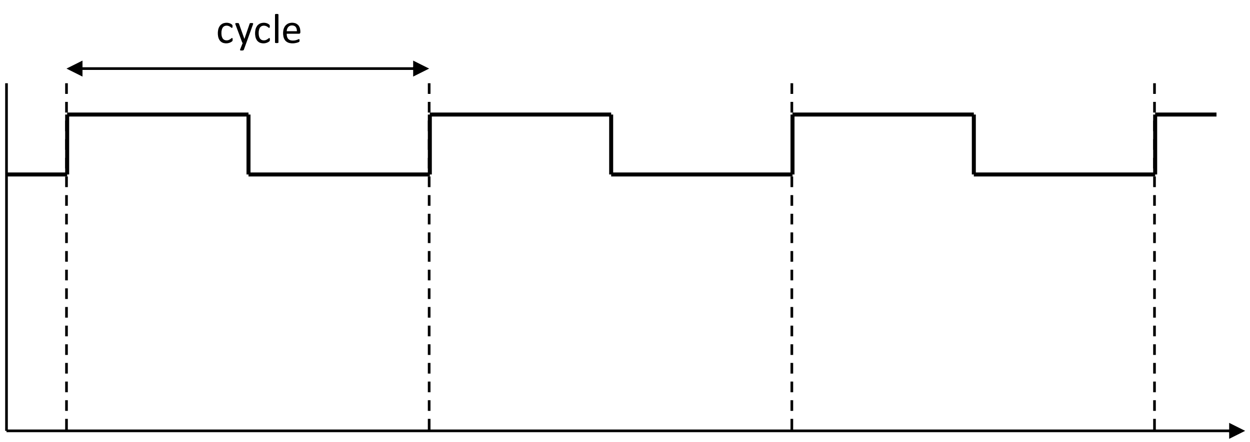
Clock Cycles

- Computer designers use a measure that relates to how fast the HW can perform basic functions
- Almost all computers are constructed using a clock that determines when events take place in HW
- These discrete time intervals are called clock cycles (*CC*)



Clock Cycle Time

- *CCT*
- Time for one *cycle*
- e.g., 250 picoseconds, or 250 ps



Clock Rate

- *CR*
- Inverse of clock cycle time
- Usually processor clock runs at a constant rate
- e.g., 4 gigahertz, or 4 GHz